



魅力统计

Fascinating Statistics

第一章 引言

Chapter 1 Introduction

Chapter1 Introduction

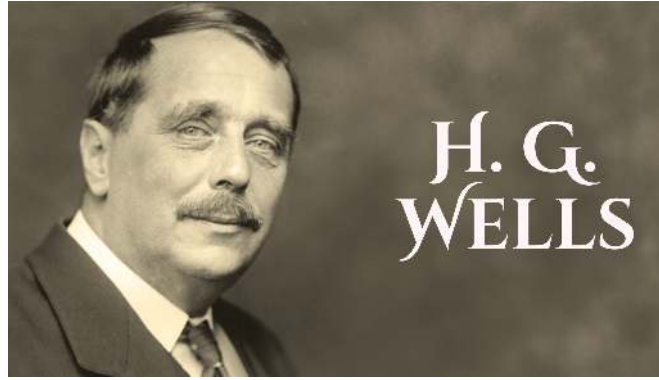
- 1.1 The Evolution of Technology
 - 1.1.1 Introduction to Statistics – The Origin of Statistics
 - 1.1.2 Introduction to Data Science – A New Era of Big Data
- 1.2 Analysis Tools



1.1.1 Introduction to Statistics



To understand God's thoughts we must study statistics, for these are the measure of His purpose.



Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write.



人工智能的本质是统计学。
The essence of artificial intelligence is statistics.



1.1.1 Introduction to Statistics

- Statistics has permeated every aspect of daily life, socio-economic activities, and scientific research; it is everywhere.



Governments can use it to conduct research and surveys in areas such as demography, macroeconomic monitoring and forecasting, social development and evaluation.



Businesses can use it for market research, customer behavior analysis, and advertising promotion.



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Pharmaceutical companies can use it to tell us whether a newly developed drug is effective in treating cancer.



Insurance companies can use it for actuarial science, risk management, etc.



1.1.1 Introduction to Statistics

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Actually, more than two-thirds of Nobel Prize in Economics achievements are related to statistics and quantitative analysis.

Two Definitions of Statistics

- The practice of describing and summarizing data or other information related to a phenomenon.
- The process of collecting, organizing, and analyzing data concerning a phenomenon.



1.1.1 Introduction to Statistics

- The earliest known statistical work is the 9th-century book *Manuscript on Deciphering Cryptographic Messages* by the Arab scholar *Al-Kindi*, which details how to use statistical data and frequency analysis for deciphering codes.

Letters	Frequency	Letters	Frequency	Letters	Frequency	Letters	Frequency
ā (ا)	600	n (ن)	221	k (ك)	112	d (د)	35
l (ل)	437	r (ر)	155	d (د)	92	ṣ (ص)	32
m (م)	320	‘ (ع)	131	s (س)	91	h (خ)	20
h (هـ)	273	f (ف)	122	q (ق)	63	t (ث)	17
(*) ū+w (و)	262	t (ت)	120	ḥ (ح)	57	ṭ (ط)	15
(*) i+y (ي)	252	b (ب)	112	ḡ (ج)	46	ḡ (غ)	15
						z (ظ)	8

A Table shewing how many died weekly, as well of all Diseases, as of the Plague, in the Years 1592, 1603, 1625, 1630, 1636; and this present Year 1665.

Buried of all Diseases in the Year 1592.		Buried of all Diseases in the Year 1603.		Buried of all Diseases in the Year 1625.		Buried of all Diseases in the Year 1630.		Buried of all Diseases in the Year 1636.		Buried of all Diseases in the Year 1665.	
Total	Pla.	Total	Pla.	Total	Pla.	Total	Pla.	Total	Pla.	Total	Pla.
March 17	210	March 17	108	March 17	263	June 24	255	April 7	119	January 3	149
March 24	131	March 24	60	March 24	216	July 1	209	July 1	235	January 10	136
March 31	219	March 31	78	March 31	243	July 8	317	July 8	217	January 17	415
April 7	307	April 7	66	April 7	319	July 15	350	July 15	250	January 24	476
April 14	203	April 14	12	April 14	256	July 22	319	July 22	219	January 31	409
April 21	292	April 21	21	April 21	311	July 29	329	July 29	277	February 7	393
April 28	310	April 28	25	April 28	305	August 5	350	August 5	350	February 14	461
May 5	350	May 5	90	May 5	393	August 12	319	August 12	319	February 21	393
May 12	319	May 12	110	May 12	311	August 19	319	August 19	319	February 28	166
May 19	300	May 19	123	May 19	379	August 26	279	August 26	279	March 7	441
May 26	450	May 26	122	May 26	401	September 2	250	September 2	250	March 14	433
June 2	410	June 2	114	June 2	395	September 9	250	September 9	250	March 21	365
June 9	441	June 9	131	June 9	424	September 16	266	September 16	266	March 28	353
June 16	399	June 16	144	June 16	310	September 23	274	September 23	274	April 4	344
June 23	301	June 23	123	June 23	340	September 30	309	September 30	309	April 11	383
June 30	310	June 30	118	June 30	340	October 7	319	October 7	319	April 18	364
July 7	460	July 7	143	July 7	422	October 14	361	October 14	361	April 25	300
July 14	410	July 14	14	July 14	378	October 21	348	October 21	348	May 2	388
July 21	441	July 21	150	July 21	410	October 28	319	October 28	319	May 9	347
July 28	350	July 28	151	July 28	381	November 4	343	November 4	343	May 16	313
August 4	350	August 4	150	August 4	417	November 11	319	November 11	319	May 23	385
August 11	350	August 11	150	August 11	381	November 18	319	November 18	319	May 30	347
August 18	350	August 18	150	August 18	381	November 25	319	November 25	319	June 6	405
August 25	350	August 25	150	August 25	381	December 2	319	December 2	319	June 13	348
September 1	350	September 1	150	September 1	381	December 9	319	December 9	319	June 20	313
September 8	350	September 8	150	September 8	381	December 16	319	December 16	319	June 27	300
September 15	350	September 15	150	September 15	381	December 23	319	December 23	319	July 4	476
September 22	350	September 22	150	September 22	381	December 30	319	December 30	319		
September 29	350	September 29	150	September 29	381	January 6	319	January 6	319		
October 6	350	October 6	150	October 6	381	January 13	319	January 13	319		
October 13	350	October 13	150	October 13	381	January 20	319	January 20	319		
October 20	350	October 20	150	October 20	381	January 27	319	January 27	319		
October 27	350	October 27	150	October 27	381	February 3	319	February 3	319		
November 3	350	November 3	150	November 3	381	February 10	319	February 10	319		
November 10	350	November 10	150	November 10	381	February 17	319	February 17	319		
November 17	350	November 17	150	November 17	381	February 24	319	February 24	319		
November 24	350	November 24	150	November 24	381	March 2	319	March 2	319		
December 1	350	December 1	150	December 1	381	March 9	319	March 9	319		
December 8	350	December 8	150	December 8	381	March 16	319	March 16	319		
December 15	350	December 15	150	December 15	381	March 23	319	March 23	319		
December 22	350	December 22	150	December 22	381	March 30	319	March 30	319		
The Total of all that have been buried is 3586		The Total of all that have been buried is 3586		The Total of all that have been buried is 3586		The Total of all that have been buried is 3586		The Total of all that have been buried is 3586		The Total of all that have been buried is 3586	
Whereof of the Plague 1159		Whereof of the Plague 1159		Whereof of the Plague 1159		Whereof of the Plague 1159		Whereof of the Plague 1159		Whereof of the Plague 1159	

- Some scholars regard the 1662 publication *Natural and Political Observations Made upon the Bills of Mortality* by *John Graunt*—compiled based on mortality tables—as marking the birth of statistics.



1.1.1 Introduction to Statistics

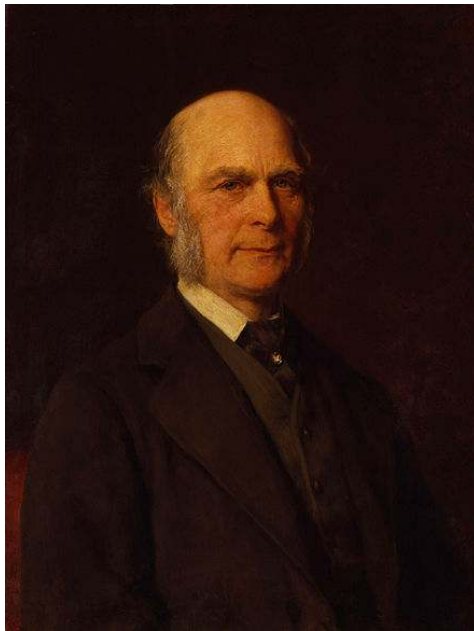
- Statistics was originally a tool of governance.
 - The term “statistics” originates from the Modern Latin *statisticum collegium* (国会).
 - In the 16th century, the Italian word *statista* referred to a statesman or someone involved in governmental affairs.
 - In 1749, the German scholar Gottfried Achenwall began using the word *Statistik* to denote the analysis of state data...
 - This eventually evolved into the modern term “statistics” which still retains the root “state” (referring to government, country, or polity).
- The practice of collecting and using data can be traced back to ancient societies, but statistics only became a systematic science a little over 300 years ago.
- In the 19th century, its integration with probability theory—the study of uncertainty—gave rise to modern statistics as we know it today.
- Statistics continues to evolve and innovate to this day, maintaining its spirit of active innovation and determined progress.



1.1.1 Introduction to Statistics

Read: <https://mp.weixin.qq.com/s/W7oksKQ1OrfWS1tFPPVSJw>

Francis Galton
弗朗西斯·高尔顿
(1822-1911)



Discovered the phenomenon of regression toward the mean and pioneered the statistical study of heredity, laying the groundwork for eugenics.

Karl Pearson
卡尔·皮尔逊
(1857-1936)



Developed regression analysis and the correlation coefficient; the first to apply rigorous mathematical methods to biostatistics.

Ronald Fisher
罗纳德·费希尔
(1890-1962)



Introduced analysis of variance, statistical hypothesis testing, and experimental design.



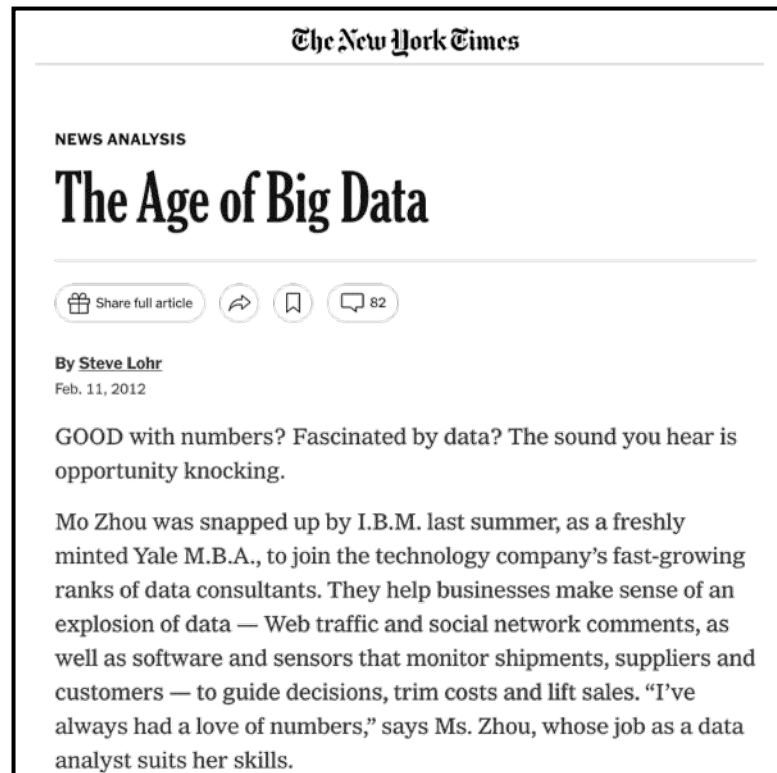
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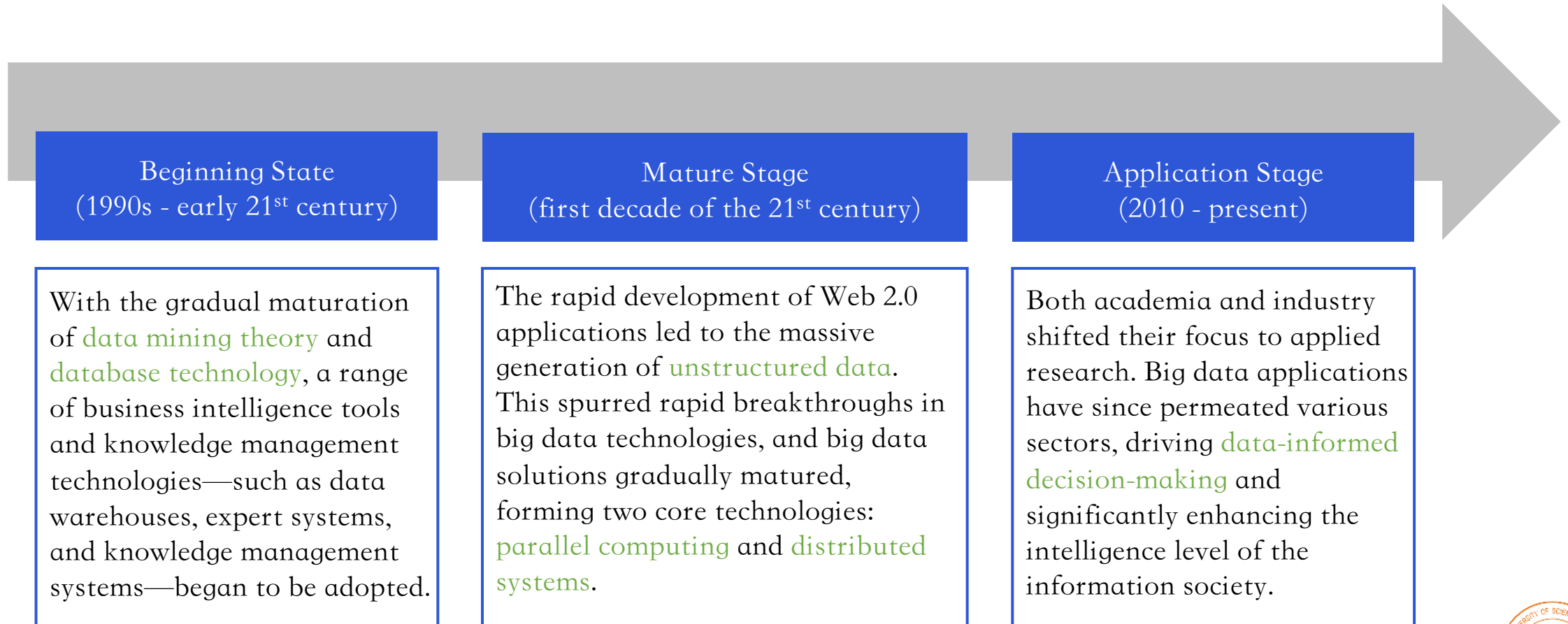
1.1.2 Introduction to Data Science

- The term "Big Data" began to emerge in the 1990s.
- In 2012, a *New York Times* column declared that "the age of big data has arrived", sparking a wave of enthusiasm around big data.
- China also referred to 2013 as the "first year of big data".



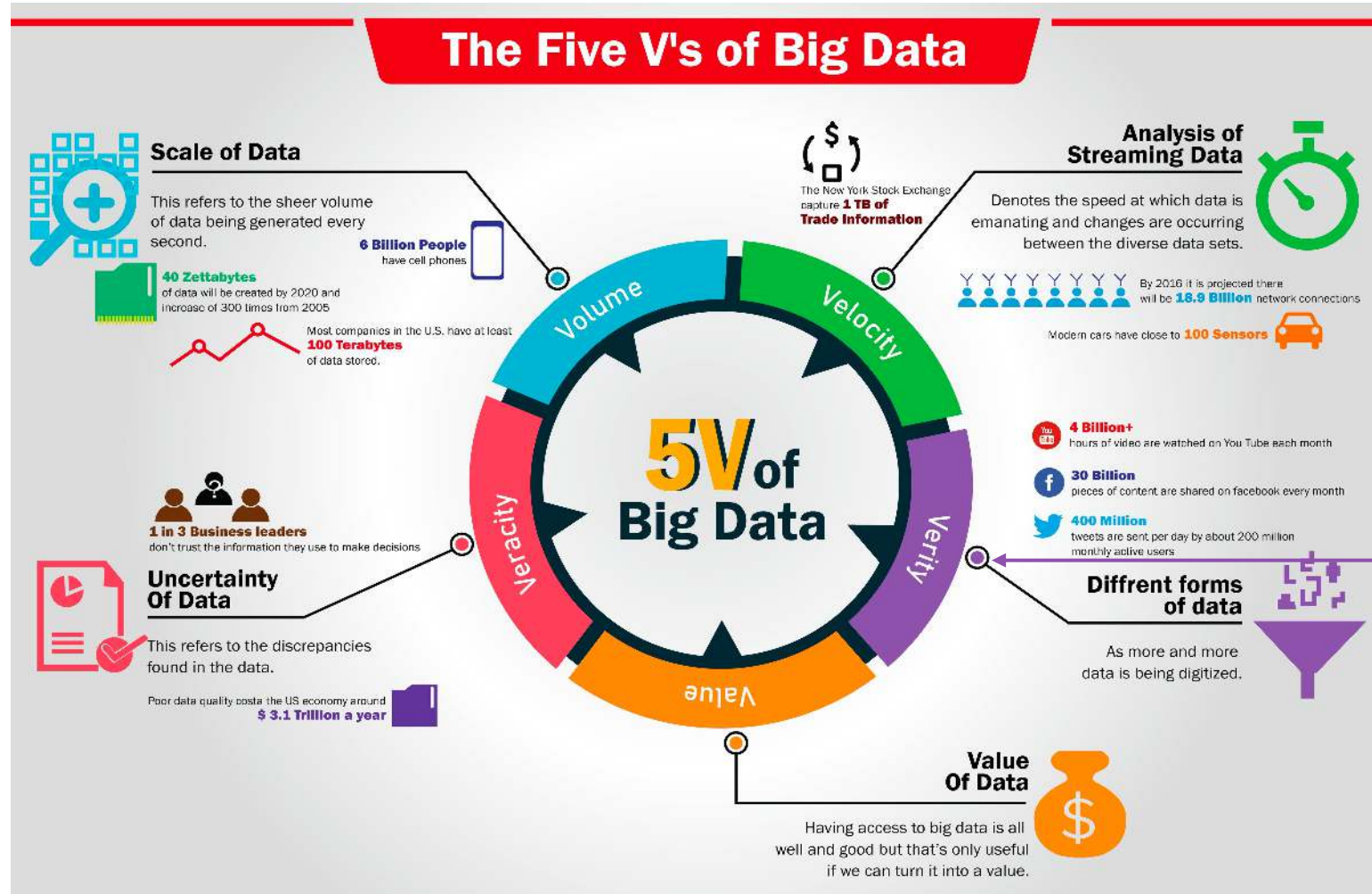
1.1.2 Introduction to Data Science

- The development history of big data:



1.1.2 Introduction to Data Science

- Big data is not only about vast volumes of data; it has been endowed with richer implications:



This should be "variety".



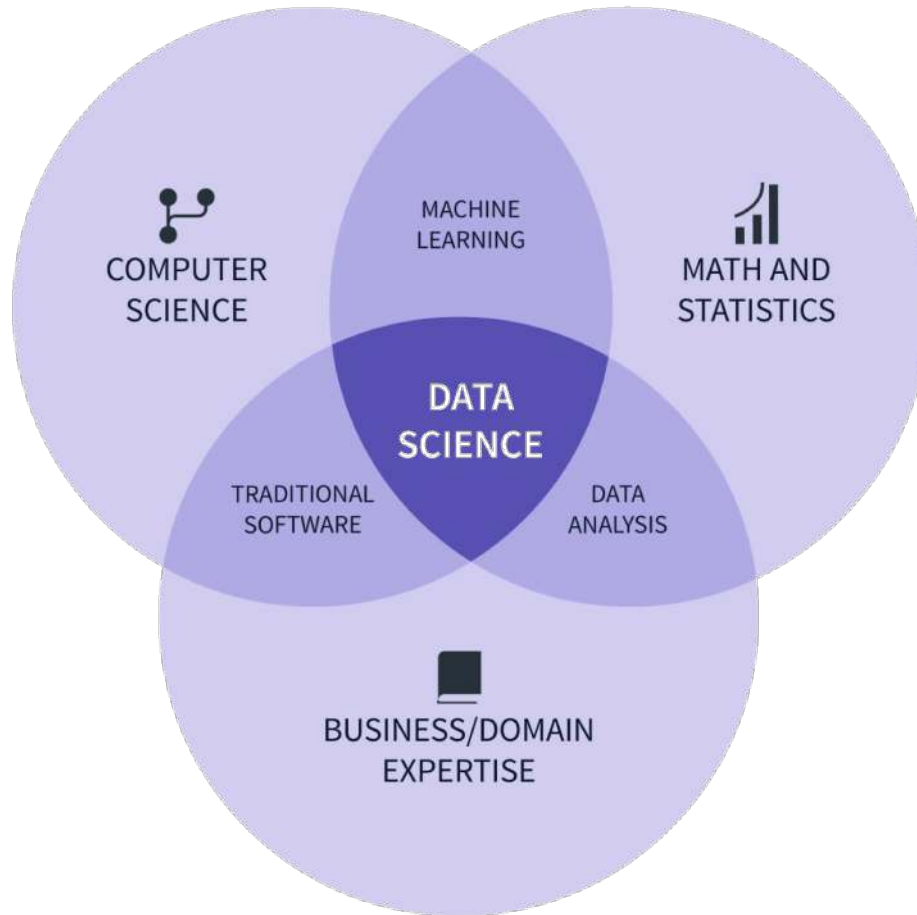
1.1.2 Introduction to Data Science

- Simply collecting large amounts of data does not automatically generate value.
- So far, there is no universal big data solution that can **automatically** perform all analyses.
- Successful application of big data in any industry or field requires tailored **analysis and modeling** based on the specific data and scenario.
- The need to analyze and model big data has made data science increasingly prominent.
- Long before the era of big data arrived, statisticians had already introduced the concept of data science:
 - **John Tukey** published the article *The Future of Data Analysis* in 1962;
 - **Jeff Wu** delivered a speech titled *Statistics = Data Science?* in 1997;
 - **William Cleveland** published the article *Data Science: An Action Plan for Expanding the Technical Areas of the Field of Statistics* in 2001.
- However, most people generally consider that data science truly gained popularity when **DJ Patil** and **Jeff Hammerbacher** defined their roles at LinkedIn and Facebook as "Data Scientist".



1.1.2 Introduction to Data Science

■ Understanding Data Science



- Data science emphasizes acquiring knowledge from data by integrating multiple theories and technical approaches based on big data.
- In terms of data storage and access, **computer technology** can be utilized to build cloud computing platforms.
- For data analysis, **statistical theory** and computing techniques can be combined to implement various high-performance analytical models.
- In the field of data application, the value of data is realized according to specific requirements, rules, and data characteristics.



1.1.2 Introduction to Data Science

Harvard
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Review

Analytics And Data Science

Data Scientist: The Sexiest Job of the 21st Century

by Thomas H. Davenport and DJ Patil

From the Magazine (October 2012)



Andrew J. Buboltz, silk screen on a page from a high school yearbook, 8.5" x 12", 2011 Tamar Cohen

Summary. Back in the 1990s, computer engineer and Wall Street “quant” were the hot occupations in business. Today data scientists are the hires firms are competing to make. As companies wrestle with unprecedented volumes and types of information, demand for these experts has raced well ahead of supply. Indeed, Greylock Partners, the VC firm that backed Facebook and LinkedIn, is so worried about the shortage of data scientists that it has a recruiting team dedicated to channeling them to the businesses in its portfolio.

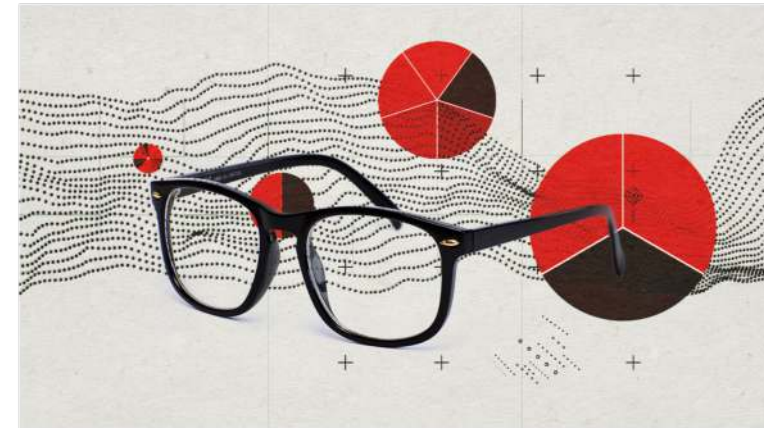
Harvard
Business
Review

Analytics And Data Science

Is Data Scientist Still the Sexiest Job of the 21st Century?

by Thomas H. Davenport and DJ Patil

July 15, 2022



HBR Staff/StudioM1/Moritz Otto/Getty Images

Summary. Ten years ago, the authors posited that being a data scientist was the “sexiest job of the 21st century.” A decade later, does the claim stand up? The job has grown in popularity and is generally well-paid, and the field is projected to experience more growth than almost any other by 2029. But the job has changed, in both large and small ways. It’s become better institutionalized, the scope of the job has been redefined, the technology it relies on has made huge strides, and the importance of non-technical expertise, such as ethics and change management, has grown. How it operates in companies — and how executives need to think



Impact of AI

Will Data analysts be fully replaced by AI in the future?

- * Short answer: Unlikely. AI will automate large chunks of a data analyst's workflow, but full replacement is improbable. The role will shift toward higher-value work that blends domain expertise, critical thinking, and communication with AI-enabled tooling.

What AI is already good at

- Data munging at scale: cleaning, joins, basic feature engineering.
- Routine analysis: descriptive stats, trend detection, simple forecasting, SQL/Python boilerplate.
- Visualization drafts: generating first-pass charts and dashboards.
- Documentation: code comments, query explanations, data dictionary summaries.

Where humans remain essential

- Problem framing: turning messy business questions into measurable hypotheses and metrics.
- Data judgment: understanding data provenance, caveats, leakage, survivorship bias, seasonality, and when results are "too good."
- Causal inference and experimentation: designing A/B tests, handling confounders, interpreting heterogeneous effects.
- Stakeholder alignment: translating insights into decisions, negotiating trade-offs, telling the story with credibility.
- Governance and ethics: privacy, fairness, compliance, and accountable use of models.
- Edge cases: novel problems and fragmented data where patterns are sparse or shifting.

Do you think that having a solid grasp of the theoretical foundation of statistics, including probability theory, statistical inference, regression analysis, etc, is still important?

- * Yes—arguably more than ever. Reasons:

- Sound decisions under uncertainty: Theory grounds concepts like variance, bias, power, and confidence/credible intervals.
- Model validity and trust: Know assumptions (iid, linearity, independence), when they're violated, and how to diagnose/fix them.
- Causality and experiments: Design A/B tests, handle confounders, interpret heterogeneous effects—beyond pattern fitting.
- Robustness and fairness: Understand overfitting, leakage, regularization, and bias-variance trade-offs to audit AI outputs.
- Interpretability and communication: Translate effects, uncertainty, and trade-offs credibly to stakeholders.
- Debugging and scaling: Choose estimators, assess identifiability, reason about asymptotics, and evaluate generalization.

What to prioritize:

- Core probability, estimation, hypothesis testing, resampling/bootstrapping.
- Linear/GLMs, causal inference basics (potential outcomes, DAGs), experimental design.
- Bayesian thinking for prior/posterior reasoning and decision-making.

Balance with practice: pair theory study with projects, simulation to test intuitions, and rigorous evaluation of models (checks, sensitivity analyses). Theory + tooling beats tooling alone.



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1.2 Analysis Tools

- The contested landscape of statistical software



SPSS

- The world's earliest statistical software;
- Developed in 1968 by Stanford University graduate student **Norman H. Nie** and others;
- Full name stands for *Statistical Package for the Social Sciences*; also means *Statistical Product and Service Solutions*;
- Acquired by IBM for 1.2 billion US dollars in July 2009.



SAS

- Arguably the world's most renowned statistical software, short for *Statistical Analysis System*;
- Developed by faculty members **Jim Goodnight** and **Jim Barr** at North Carolina State University;
- In 1976, SAS Institute Inc. was officially founded and has since grown into one of the world's largest software companies.
- Its client base includes government agencies, banks, pharmaceutical companies, etc.



1.2 Analysis Tools

- The favorite of statisticians



R & RStudio

- R can be considered a "dialect" of the S language.
- The S language was created in the Statistics Research Department at Bell Labs, led by John M. Chambers. In 1998, it was awarded the **ACM Software System Award**—the only statistical software or language to have received this honor to date.
- In 1992, Ross Ihaka and Robert Gentleman at the University of Auckland developed a new language based on S for teaching purposes, naming it "R" after the first letters of their names.
- R is free and open-source software, now maintained by the R Core Team, though users worldwide can contribute and share packages.
- RStudio is an IDE for R. It integrates many powerful features—such as a code editor, auto-completion, graphics visualization, and data management—into an intuitive and easy-to-learn interface, making R programming more convenient and efficient.



1.2 Analysis Tools

- The favorite of data scientists



Python

- The programming language Python was created in 1989 by Dutch programmer Guido van Rossum.
- The name "Python" was chosen not for its literal meaning (the snake), but simply because he was a fan of the comedy group *Monty Python*.
- Python is an open-source programming language with a highly open and active community.
- Its rise as a mainstream language in data science is due not only to its programming flexibility, but to its **rich ecosystem**: nearly all the latest machine learning and AI algorithms can quickly be implemented via Python packages, and most open-source projects and new computational frameworks offer Python interfaces.



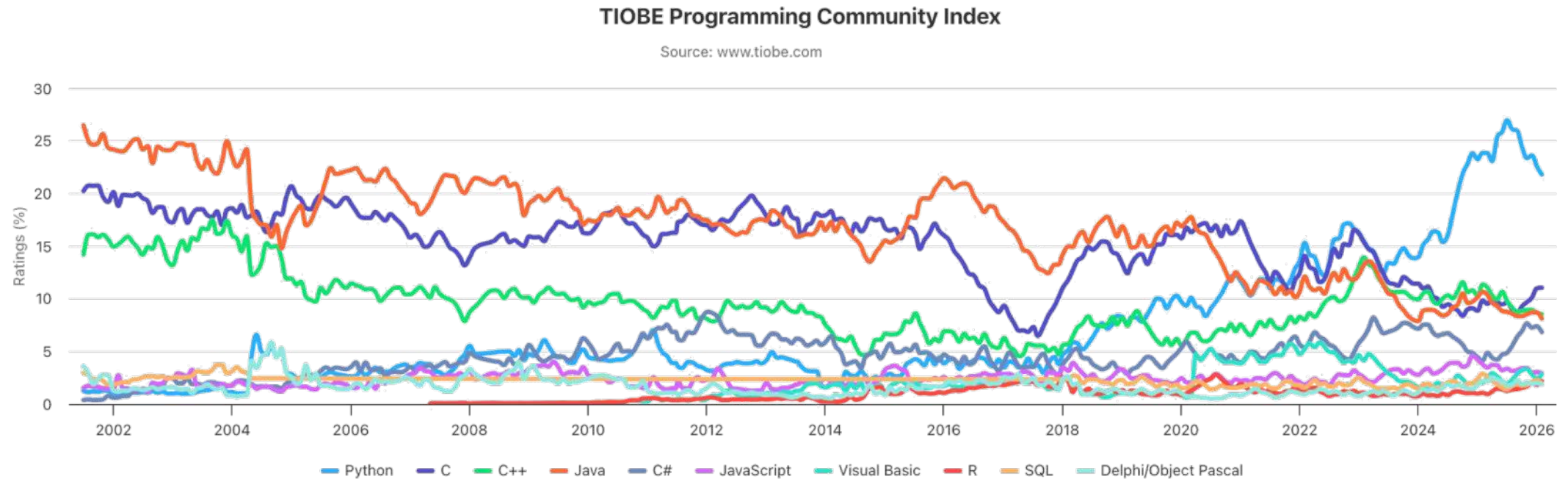
1.2 Analysis Tools

- Python, one of today's most popular programming languages, features concise and elegant syntax that is nearly executable pseudocode.
- The most effective way to learn Python is by getting your hands dirty. Type out all the code in the link below; that's all you really need to get the fundamentals down:
 - English: <https://learnxinyminutes.com/python/>
 - Chinese: <https://learnxinyminutes.com/zh-cn/python/>
- Install Python and Jupyter Notebook: (you can find a vast number of tutorials online)
 - Python: <https://www.geeksforgeeks.org/python/download-and-install-python-3-latest-version/>
 - Jupyter Notebook: <https://www.dataquest.io/blog/jupyter-notebook-tutorial/>
- Don't worry, Python will be used solely for the purpose of demonstrating examples in class and for a limited number of homework problems. **You will not be assessed on it in the examinations.**



1.2 Analysis Tools

- Analytical tools are constantly evolving.
- There's no need to fixate on which specific programming language to learn—the underlying logic of programming is largely universal.



Source: <https://www.tiobe.com/tiobe-index/>



1.2 Analysis Tools

- Moreover, with the powerful assistance of **AI tools** in programming today, getting started and writing effective code has become much more accessible.

CS146S: The Modern Software Developer

Stanford University • Fall 2025



Course Description

In the last few years, large language models have introduced a revolutionary new paradigm in software development. The traditional software development lifecycle is being transformed by AI automation at every stage, raising the question: how should the next generation of software engineers leverage these advances to 10x their productivity and prepare for their careers?

This course demonstrates that modern AI tooling will not only enhance developer productivity but also democratize software engineering for a broader audience. We'll show that software development has evolved from 0-1 code creation to an iterative workflow of plan, generate with AI, modify, and repeat. Students will master both the theory behind traditional software engineering challenges and the cutting-edge AI-powered tools solving them today.

Through hands-on engineering tasks and talks from industry pioneers building these revolutionary tools, you'll gain practical experience with AI-assisted development, automated testing, intelligent documentation, and security vulnerability detection. By the end of this course, you'll have a crisp understanding of how to integrate state-of-the-art LLM models into complex development workflows and avoid common pitfalls.

Source: <https://themodernsoftware.dev/>



